



Day 02 — Data Visualization

Base R vs. ggplot2, the grammar of graphics, scatter/line/bar/pie plots, boxplots/histograms/density, color scales, groups, and faceting

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Package management

Same as every session: bootstrap pacman, then load the tidyverse.

```
if (!require(pacman)) install.packages("pacman")
```

MESSAGE

Loading required package: pacman

```
p_load(tidyverse)
```

Getting the data

German federal election polling trend (ZEIT ONLINE), one row per poll x party. Already fetched and reshaped to `polls.csv`, checked into this folder, so you can work offline.

```
# this is how it was fetched and reshaped:
# p_load(jsonlite)
# fromJSON("https://interactive.zeit.de/g/cronjobs/wahltrend-2025/bund/polls.json") |>
#   as_tibble() |>
#   pivot_longer(CDUCSU:BSW, names_to = "party", values_to = "share") |>
#   write_csv("d02/polls.csv")
# (Day 3 picks this reshaping step up properly, starting from the wide file.)

polls <- read_csv("d02/polls.csv")
```

MESSAGE

Rows: 14539 Columns: 7
— Column specification —
Delimiter: ","
chr (4): Poll_ID, Pollster, Period, party
dbl (2): n, share
date (1): Date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
polls |> head(100)
```

➤ OUTPUT

```
# A tibble: 100 × 7
  Poll_ID Pollster Date n Period party share
  <chr> <chr> <date> <dbl> <chr> <chr> <dbl>
1 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. CDU/CSU 37
2 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. SPD 21
3 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. Gruene 10
4 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. FDP 6
5 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. Linke 9
6 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. AfD 12
7 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. BSW NA
8 BT01170105P Infratest dimap 2017-01-05 1505 02.01.-04.01. CDU/CSU 37
9 BT01170105P Infratest dimap 2017-01-05 1505 02.01.-04.01. SPD 20
10 BT01170105P Infratest dimap 2017-01-05 1505 02.01.-04.01. Gruene 9
# i 90 more rows
```

Base R: a quick look

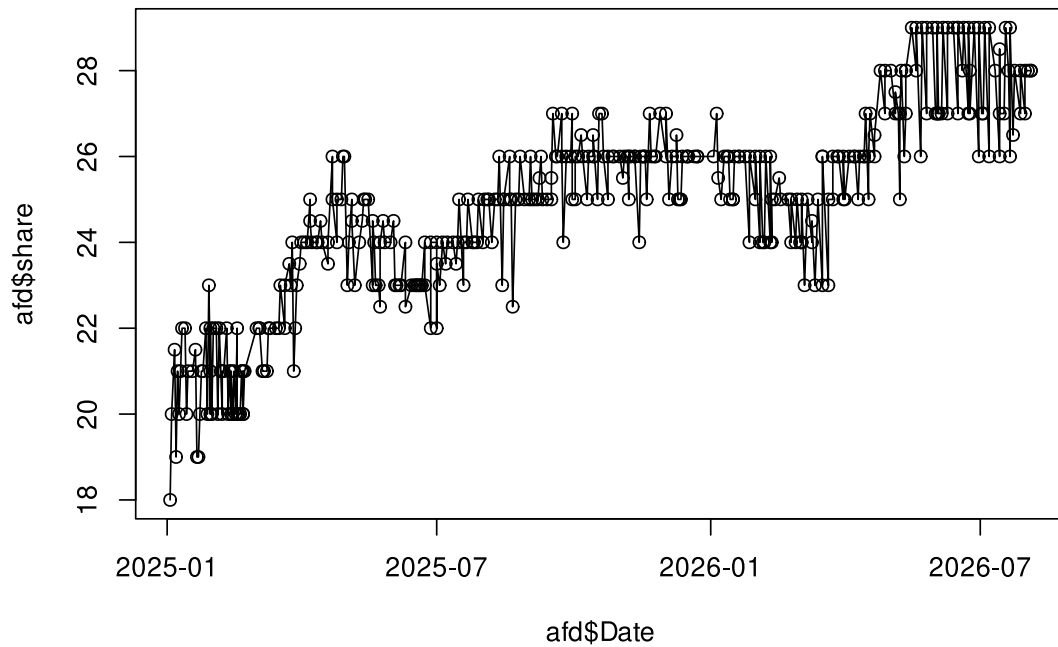
Two of the most common plots, quickly, before we move on to ggplot2.

```
afd <- polls |>
  filter(party == "AfD") |>
  filter(Date >= as.Date("2025-01-01"))
afd
```

➤ OUTPUT

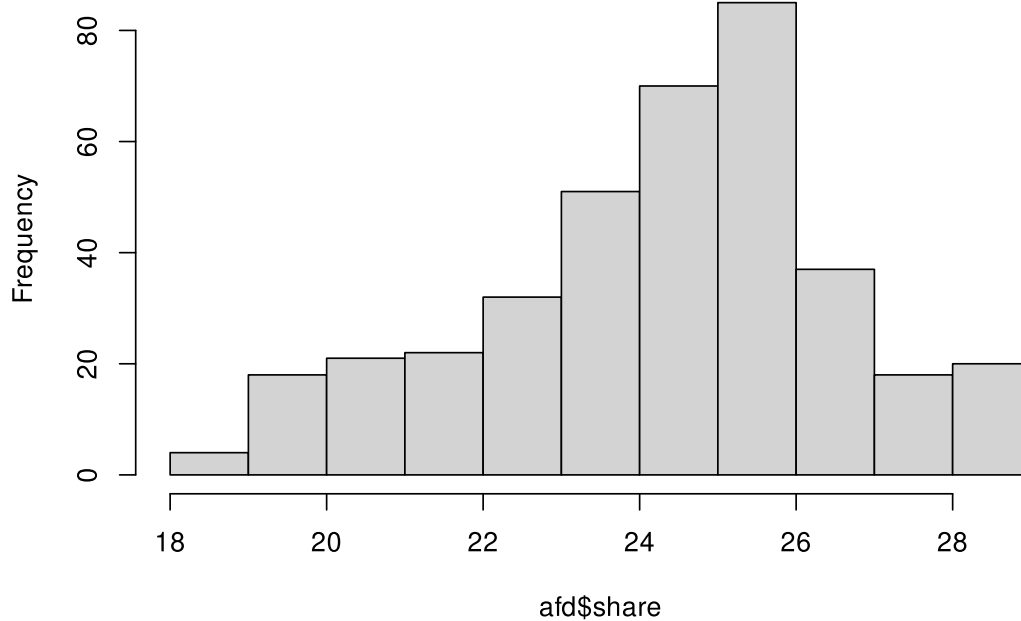
```
# A tibble: 378 × 7
  Poll_ID Pollster Date n Period party share
  <chr> <chr> <date> <dbl> <chr> <chr> <dbl>
1 BT04250103X GMS 2025-01-03 1010 27.12.-02.0... AfD 18
2 BT05250104V INSA 2025-01-04 1201 30.12.-03.0... AfD 20
3 BT05250106W INSA 2025-01-06 2001 03.01.-06.0... AfD 21.5
4 BT02250107K Forsa 2025-01-07 1501 02.01.-06.0... AfD 19
5 BT13250108Y YouGov 2025-01-08 1908 03.01.-06.0... AfD 21
6 BT01250109P Infratest dimap 2025-01-09 1323 06.01.-08.0... AfD 20
7 BT06250110D Forschungsgruppe Wahlen 2025-01-10 1433 07.01.-09.0... AfD 21
8 BT05250111X INSA 2025-01-11 1205 06.01.-10.0... AfD 22
9 BT05250113Y INSA 2025-01-13 2005 10.01.-13.0... AfD 22
10 BT02250114L Forsa 2025-01-14 2504 07.01.-13.0... AfD 20
# i 368 more rows
```

```
plot(afd$Date, afd$share)
lines(afd$Date, afd$share)
```



```
hist(afd$share)
```

Histogram of afd\$share



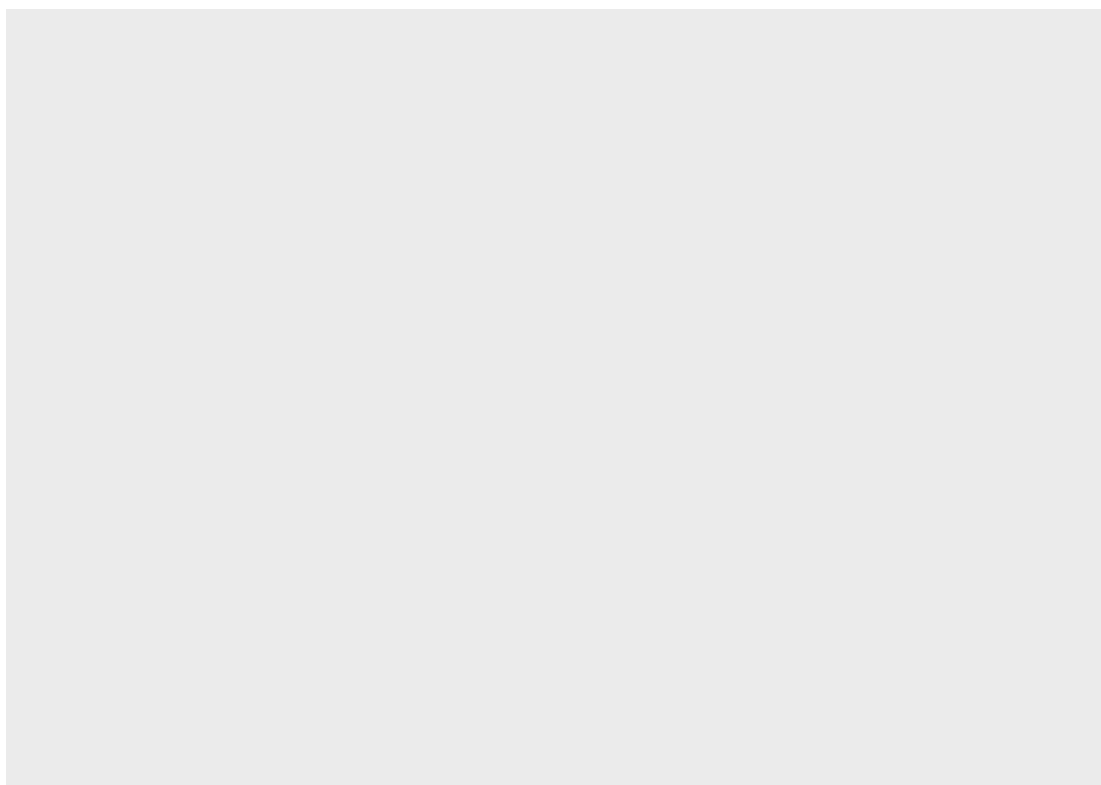
Why not base R?

Every `plot()` call starts from a blank canvas — no axis labels worth reading by default, no legend, no way to color by group without writing it by hand. `ggplot2` builds a plot in layers instead, so from here on we use it exclusively.

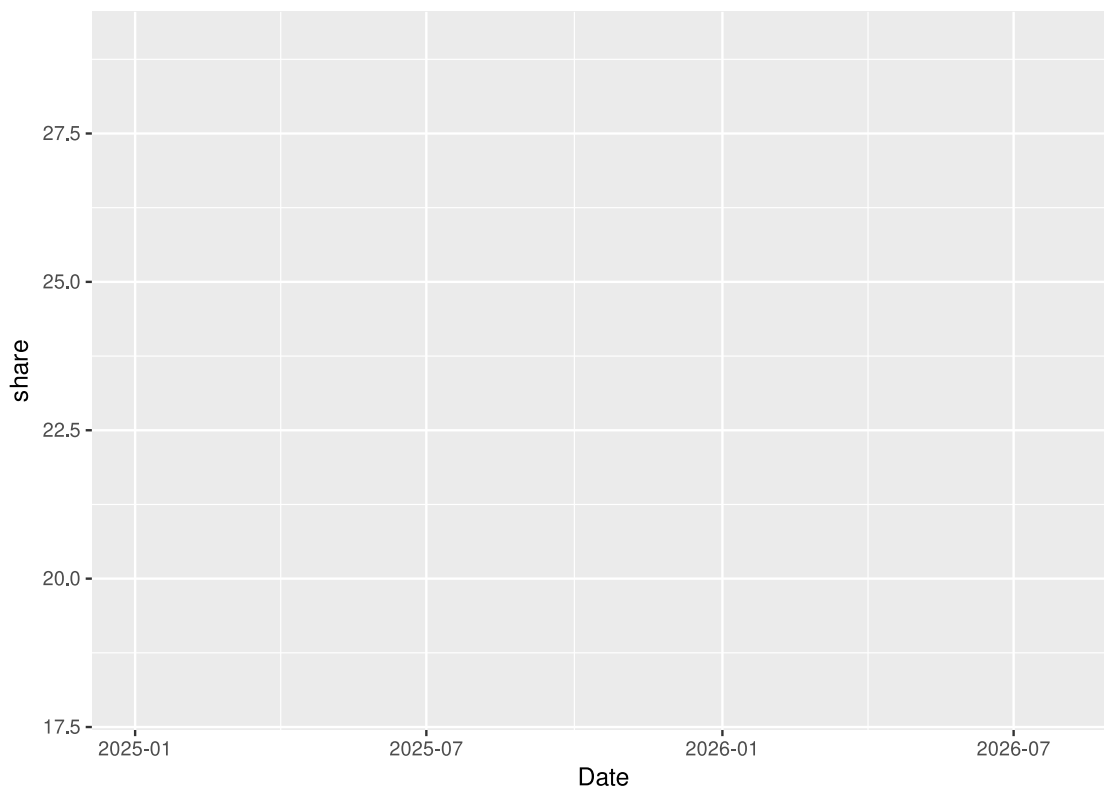
The grammar of graphics

Every ggplot starts from a blank canvas, built up step by step with `+`.

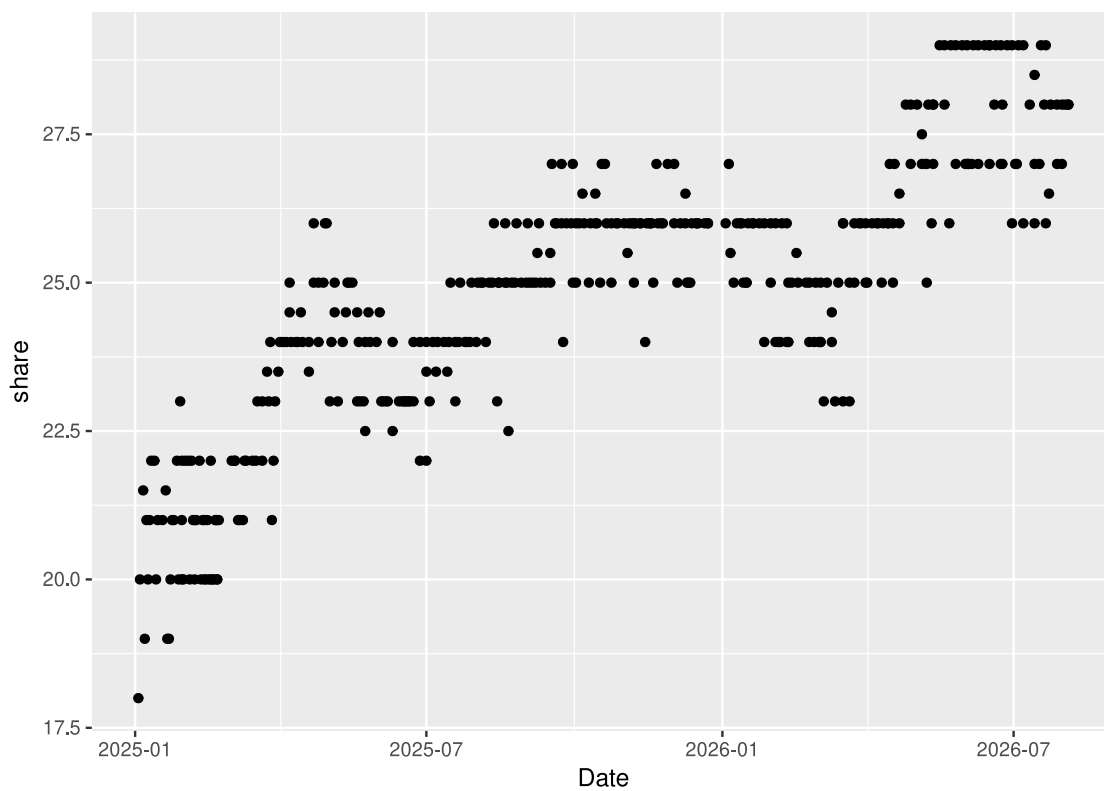
```
ggplot(polls)
```



```
ggplot(afd, aes(x = Date, y = share))
```



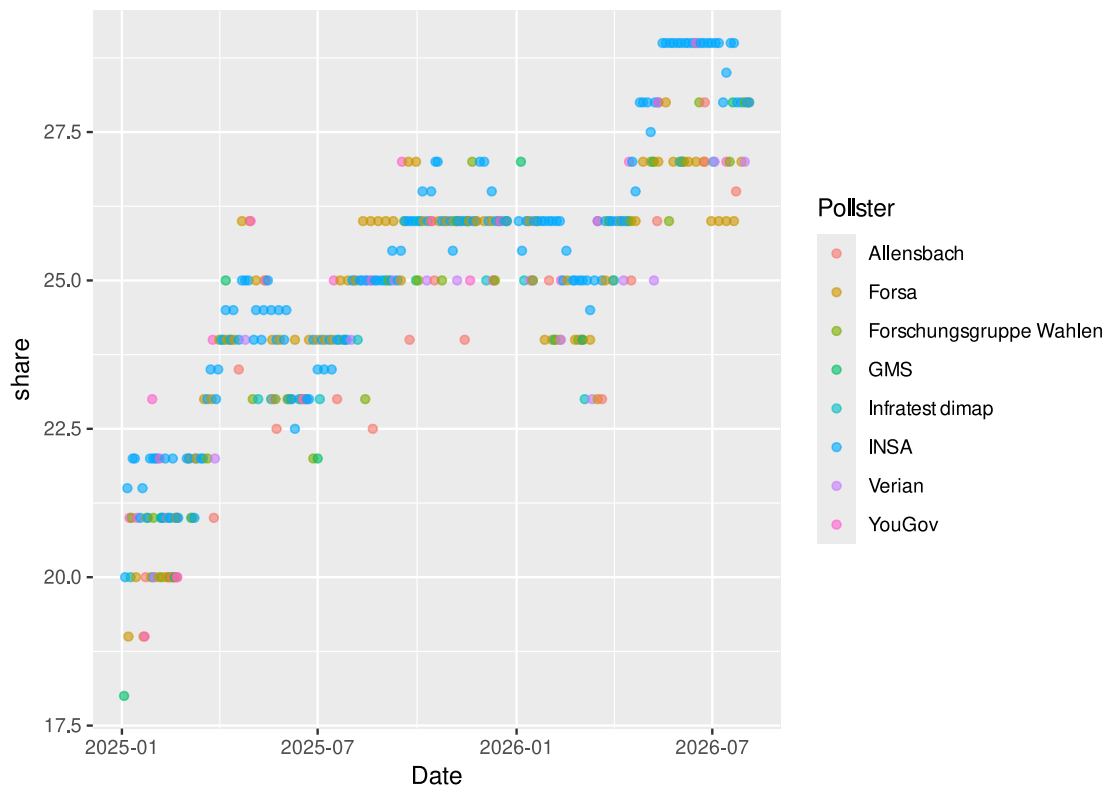
```
afd |>  
  ggplot(aes(x = Date, y = share)) +  
  geom_point()
```



Scatter plots

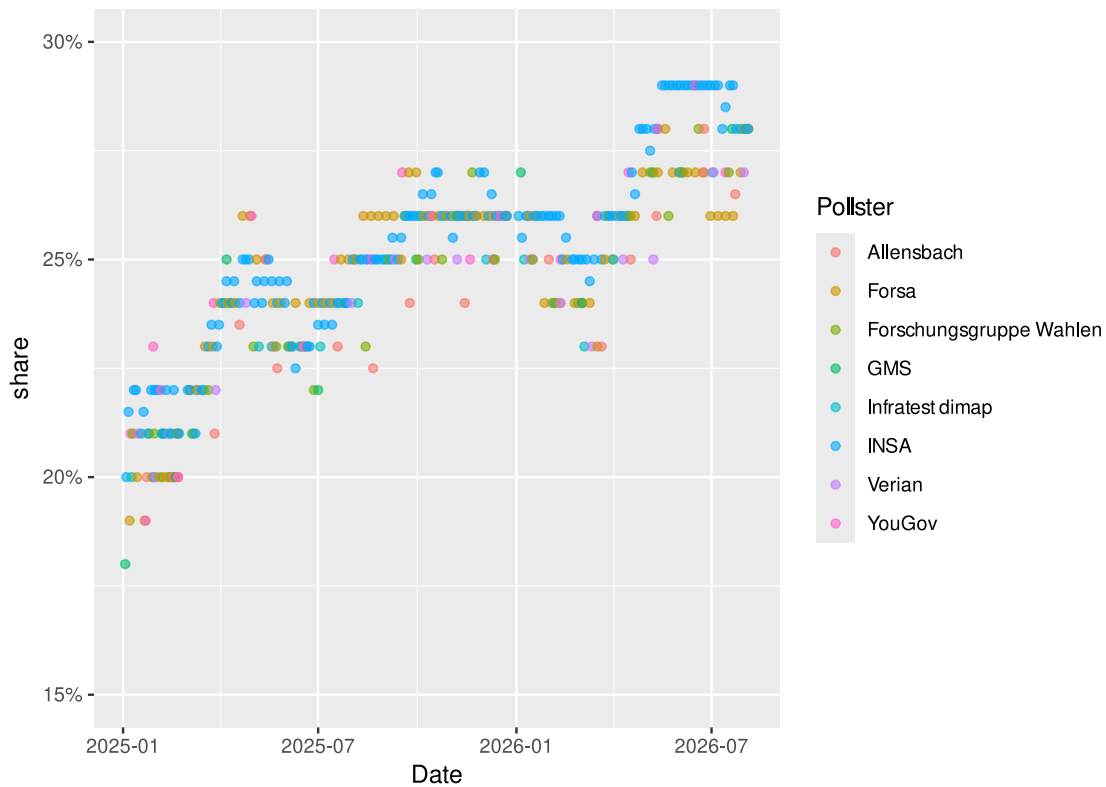
One point per poll, colored by pollster.

```
afd |>
  ggplot(aes(x = Date, y = share, color = Pollster)) +
  geom_point(alpha = 0.6)
```



Add labels, change axis breaks, and set limits.

```
afd |>
  ggplot(aes(x = Date, y = share, color = Pollster)) +
  geom_point(alpha = 0.6) +
  scale_y_continuous(
    breaks = seq(15, 30, by = 5),
    labels = c("15%", "20%", "25%", "30%"),
    limits = c(15, 30)
  )
)
```



A handy shortcut: format the y-axis as percent directly (share needs dividing by 100 first).


```
p_load(scales)
```

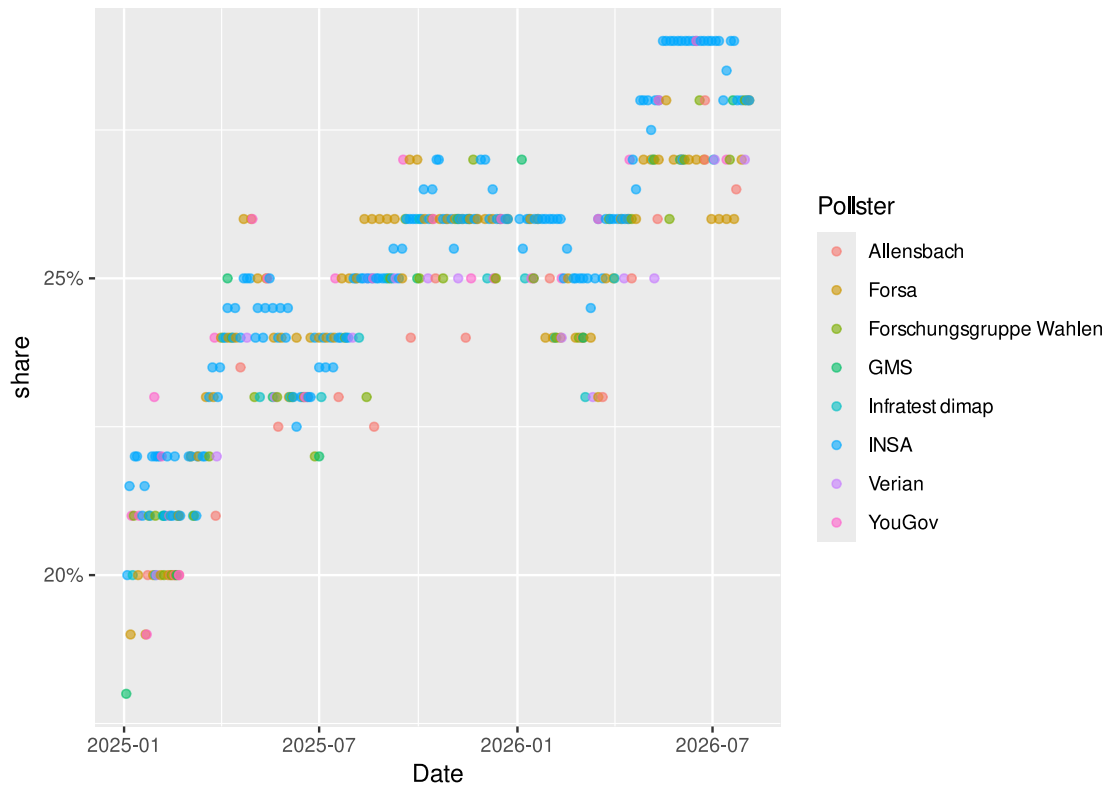
```
afd |>
```

```
  mutate(share = share / 100) |>
```

```
  ggplot(aes(x = Date, y = share, color = Pollster)) +
```

```
  geom_point(alpha = 0.6) +
```

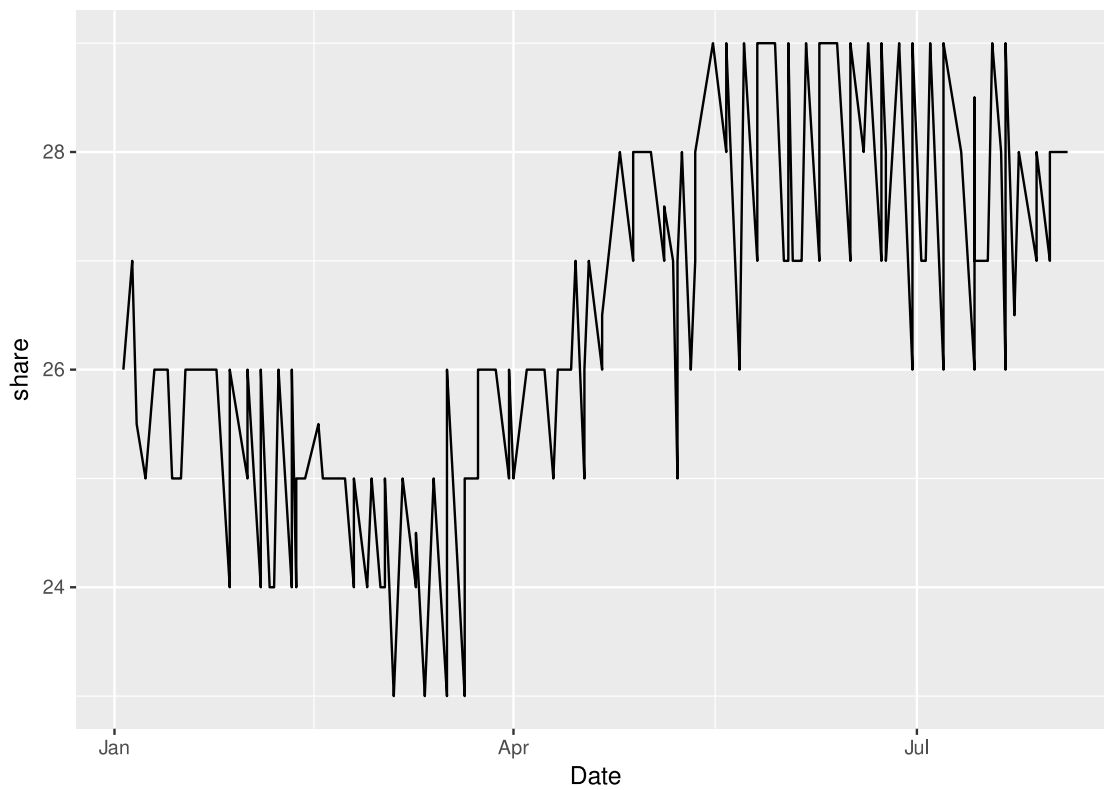
```
  scale_y_continuous(breaks = breaks_width(.05), labels = label_percent())
```



Line plots

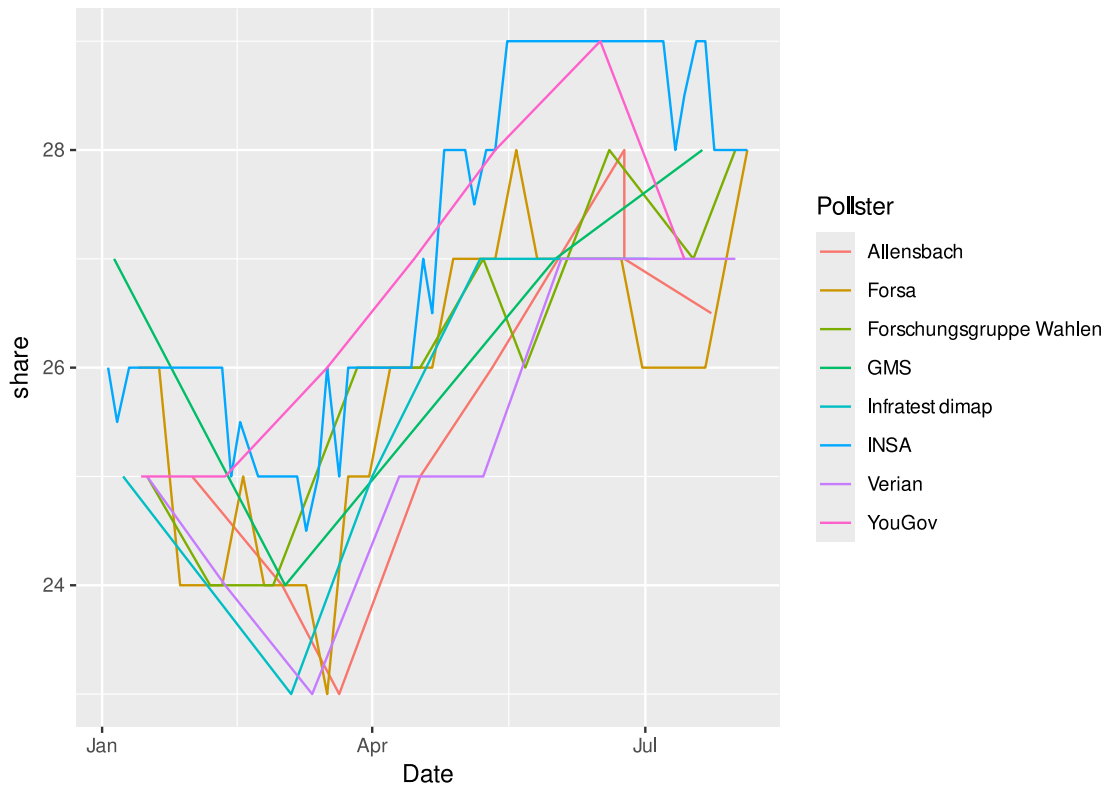
Connecting raw polls in order is noisy with this much day-to-day variation — we reduce the plotted timeframe by filtering first.

```
afd |>
  filter(between(Date, as.Date("2026-01-01"), Sys.Date())) |>
  ggplot(aes(x = Date, y = share)) +
  geom_line()
```



A line plot makes more sense per pollster.

```
afd |>
  filter(between(Date, as.Date("2026-01-01"), Sys.Date())) |>
  ggplot(aes(x = Date, y = share, color = Pollster)) +
  geom_line()
```

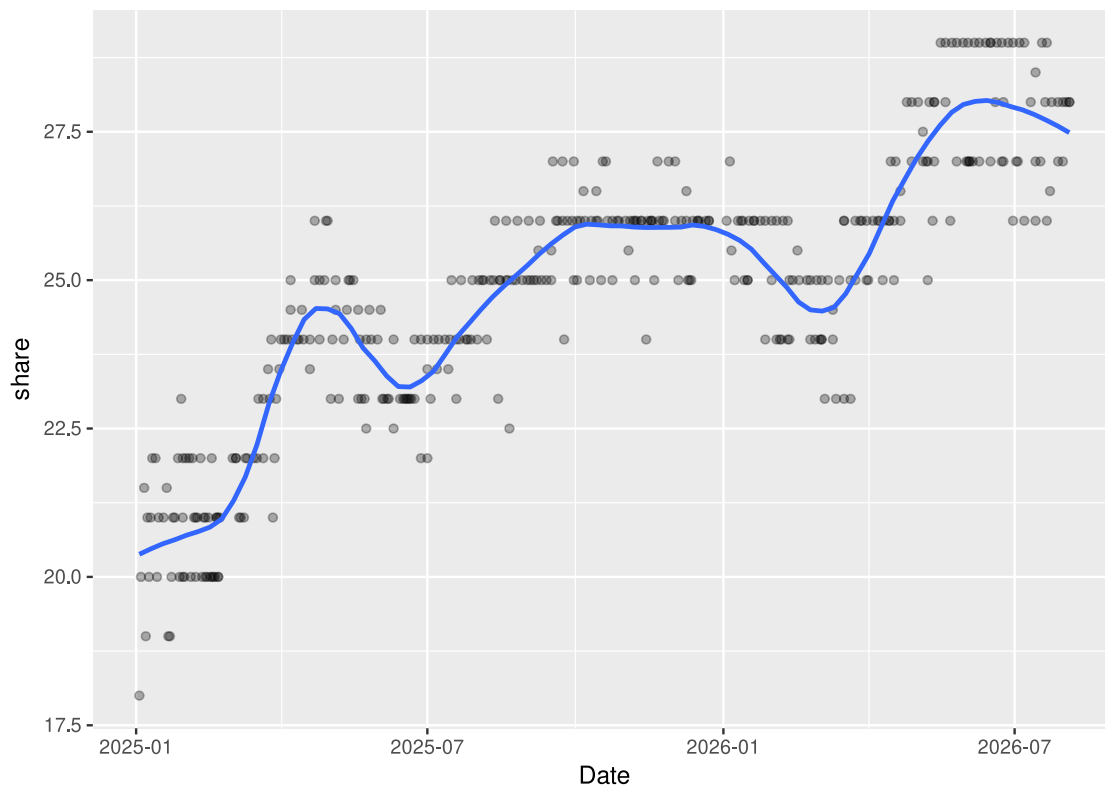


A smoothed trend line reads much better — roughly what a “polling average” tracker shows (their exact weighting scheme is more involved than this).

```
afd |>
  ggplot(aes(x = Date, y = share)) +
  geom_point(alpha = 0.3) +
  geom_smooth(span = 0.2, se = FALSE)
```

MESSAGE

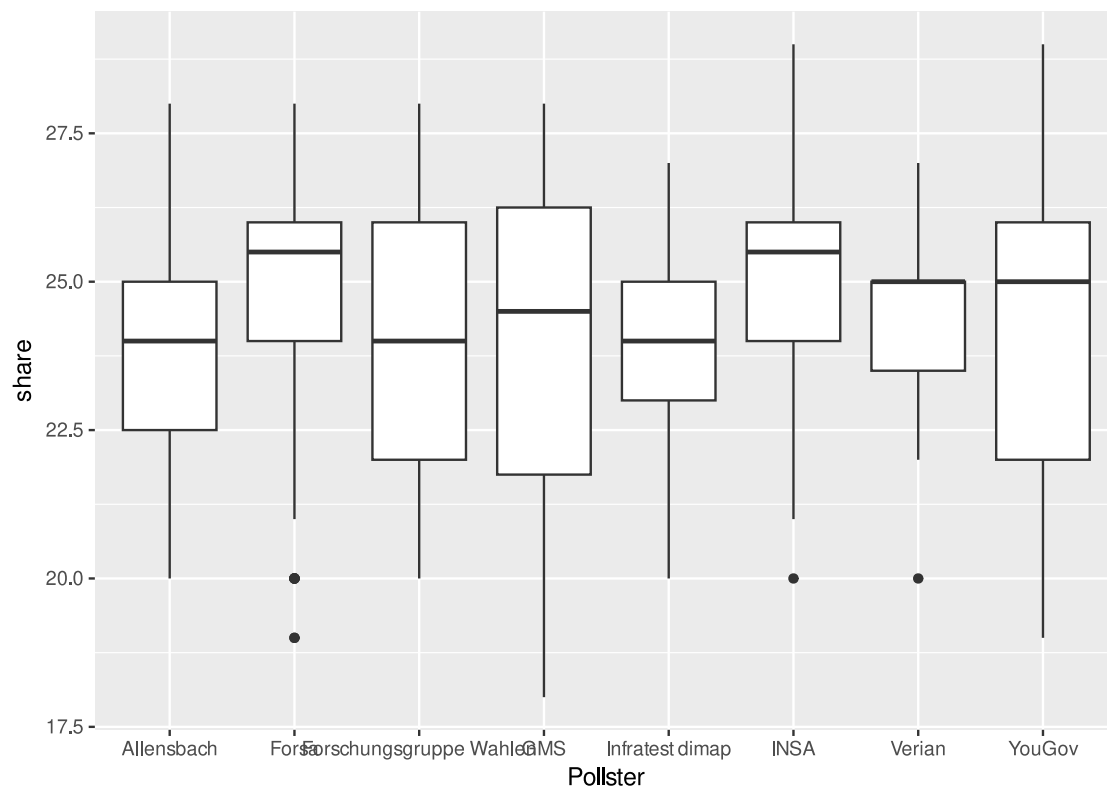
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



Boxplots

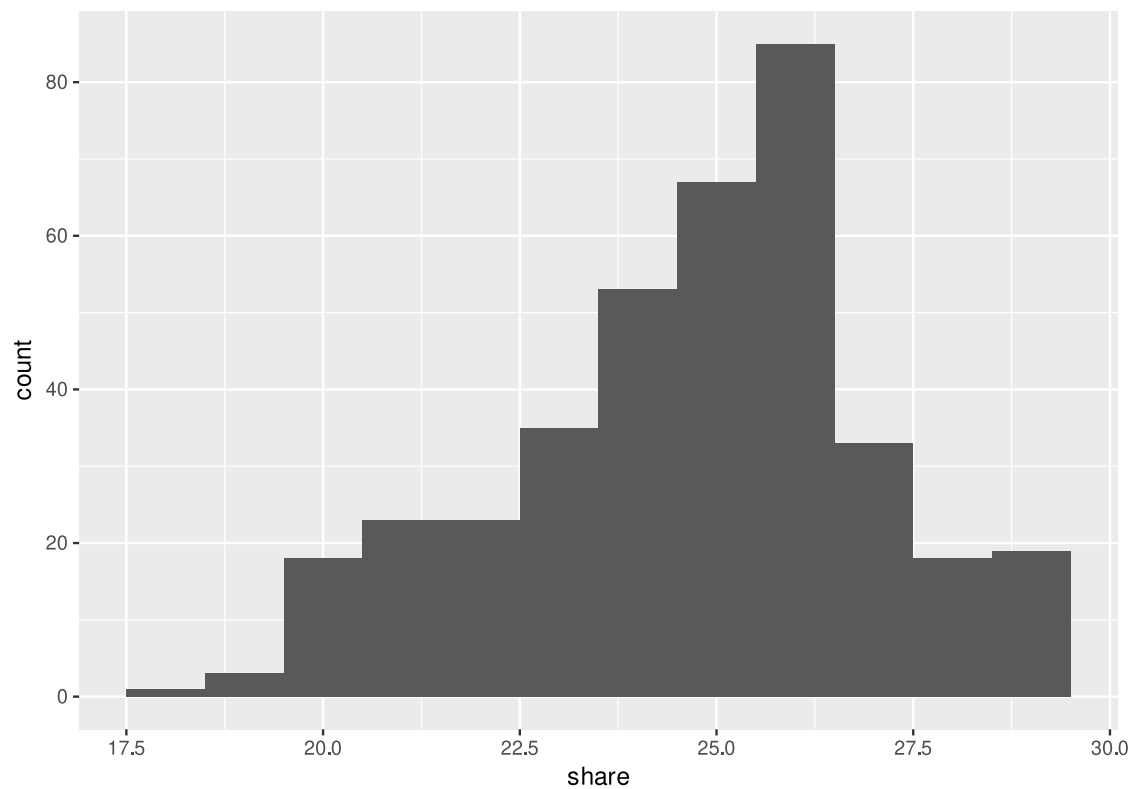
Spread of a party's polled share per pollster — “house effects”.

```
afd |>
  ggplot(aes(x = Pollster, y = share)) +
  geom_boxplot()
```



Histogram

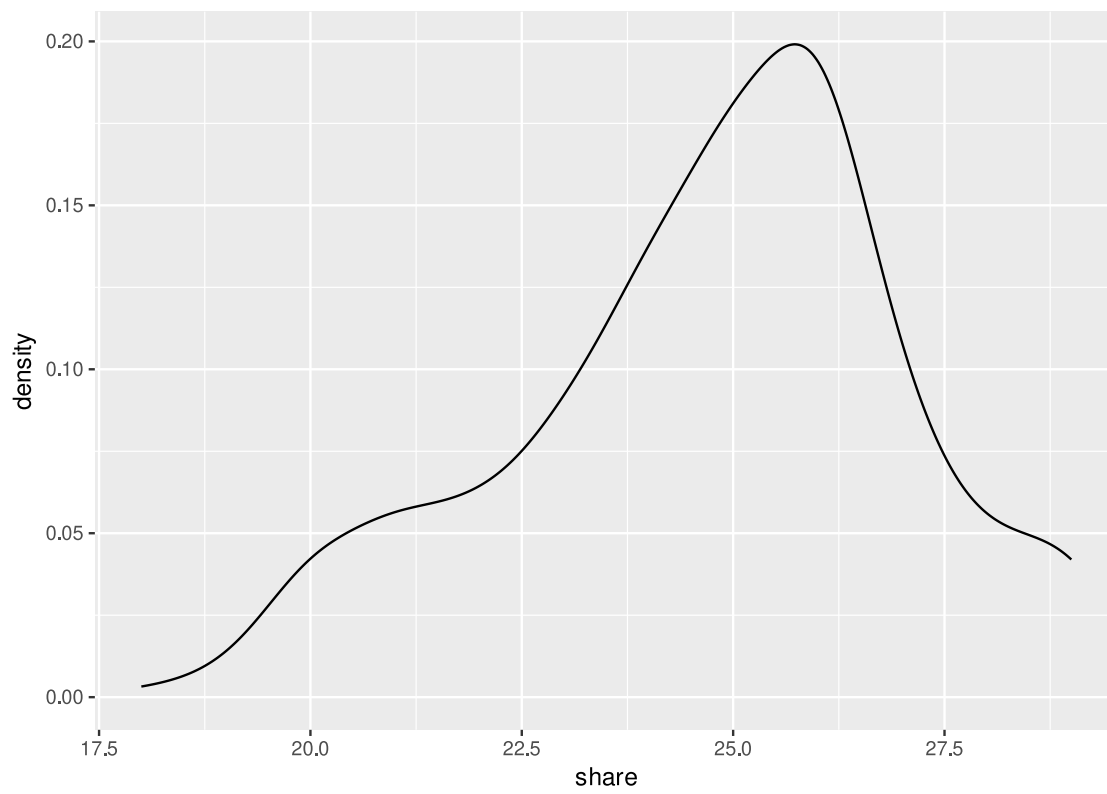
```
afd |>  
  ggplot(aes(x = share)) +  
  geom_histogram(binwidth = 1)
```



Density plot

A smoothed version of the same distribution.

```
afd |>
  ggplot(aes(x = share)) +
  geom_density()
```



Color scales

Real party colors instead of ggplot's defaults.

```
party_colors <- c(
  CDU = "#000000",
  SPD  = "#E3000F",
  Gruene = "#46962b",
  FDP   = "#ffed00",
  Linke  = "#be3075",
  AfD    = "#009ee0",
  BSW    = "#7d3f98"
)
party_colors
```

➤ OUTPUT

CDU	SPD	Gruene	FDP	Linke	AfD	BSW
"#000000"	"#E3000F"	"#46962b"	"#ffed00"	"#be3075"	"#009ee0"	"#7d3f98"

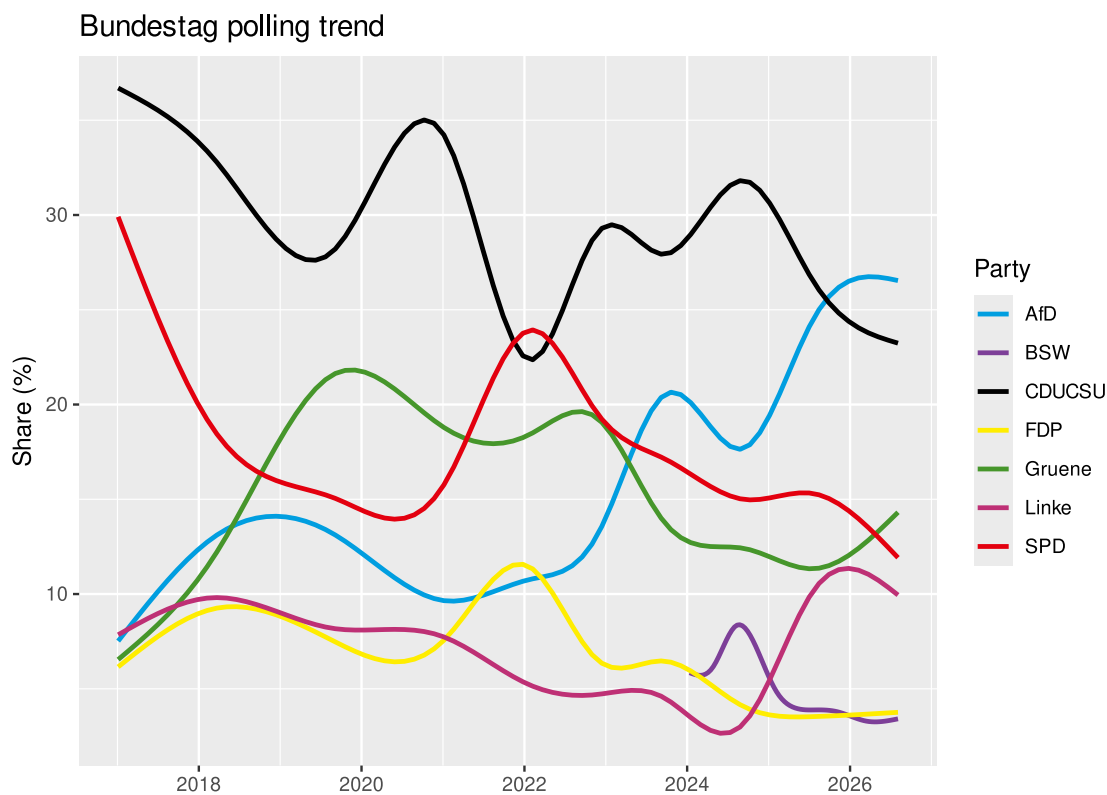
```
polls |>
  ggplot(aes(x = Date, y = share, color = party)) +
  geom_smooth(se = FALSE) +
  scale_color_manual(values = party_colors) +
  labs(title = "Bundestag polling trend", x = NULL, y = "Share (%)", color = "Party")
```

MESSAGE

`geom\_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

WARNING

Warning: Removed 1548 rows containing non-finite outside the scale range (`stat\_smooth()`).



Saving plots

`ggsave()` saves the last plot drawn, or a specific one via `plot = ....width/height` set the physical size, `dpi` the pixel density — dpi only matters for raster formats like PNG, a PDF is a vector graphic with no fixed resolution.


```
trend_plot <- polls |>
  ggplot(aes(x = Date, y = share, color = party)) +
  geom_smooth(se = FALSE) +
  scale_color_manual(values = party_colors) +
  labs(title = "Bundestag polling trend", x = NULL, y = "Share (%)", color = "Party")

ggsave("afd_trend.png", plot = trend_plot, width = 7, height = 5, dpi = 300)
ggsave("afd_trend.pdf", plot = trend_plot, width = 7, height = 5)
```

Bar plots

One bar per party, from the most recent Forsa poll.

```
latest_forsa <- polls |>
  filter(Pollster == "Forsa", Date == max(Date))
latest_forsa
```

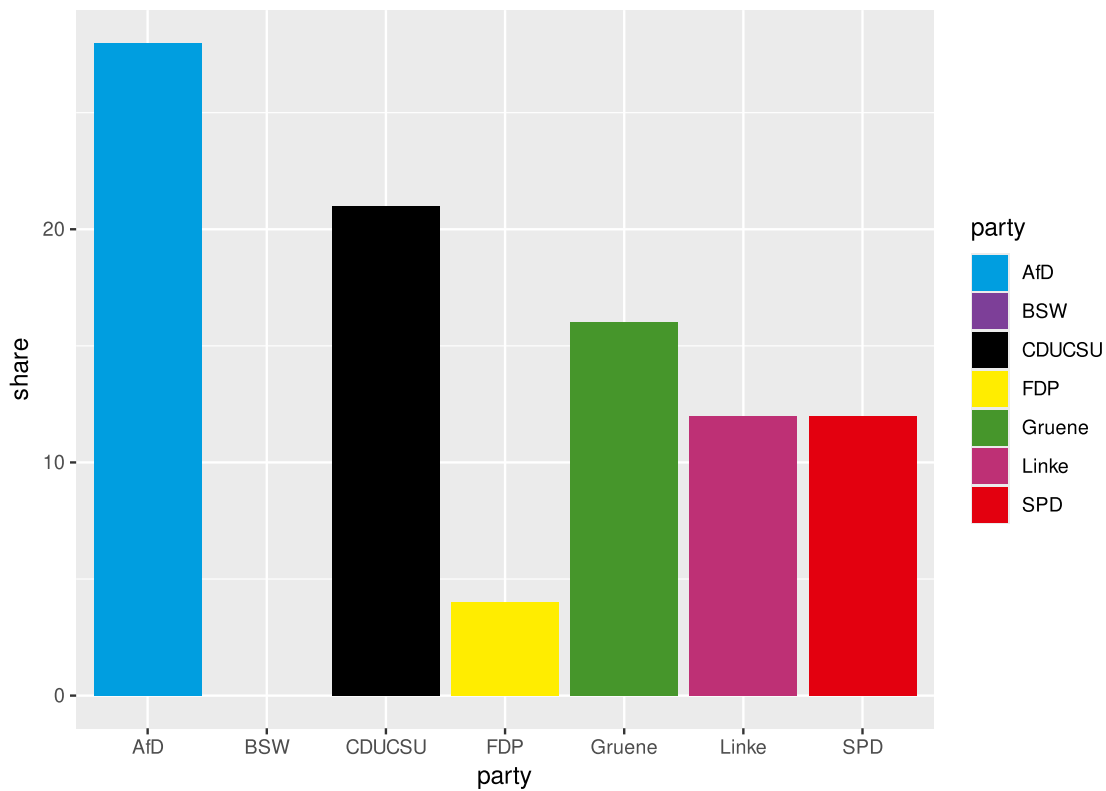
➤ OUTPUT

```
# A tibble: 7 × 7
  Poll_ID Pollster Date          n Period      party share
  <chr>    <chr>   <date>    <dbl> <chr>      <chr> <dbl>
1 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. CDU/CSU 21
2 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. SPD      12
3 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. Gruene 16
4 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. FDP      4
5 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. Linke 12
6 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. AfD     28
7 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. BSW     NA
```

```
latest_forsa |>
  ggplot(aes(x = party, y = share, fill = party)) +
  scale_fill_manual(values = party_colors) +
  geom_col()
```

⚠ WARNING

Warning: Removed 1 row containing missing values or values outside the scale range (`geom\_col()`).



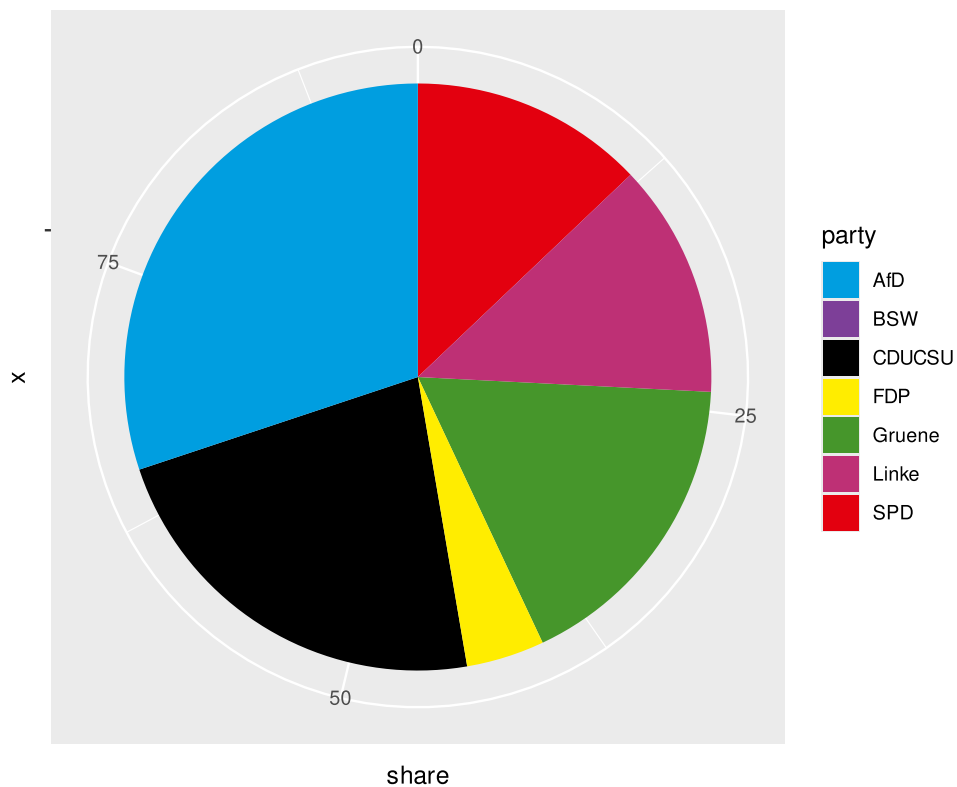
Pie plots

ggplot2 has no dedicated pie geom — it is a bar chart bent into a circle.

```
latest_forsa |>
  ggplot(aes(x = "", y = share, fill = party)) +
  geom_col() +
  coord_polar("y") +
  scale_fill_manual(values = party_colors)
```

⚠ WARNING

Warning: Removed 1 row containing missing values or values outside the scale range (`geom\_col()`).



Angles are harder to compare at a glance than bar height — the bar chart above communicates the same numbers more clearly.

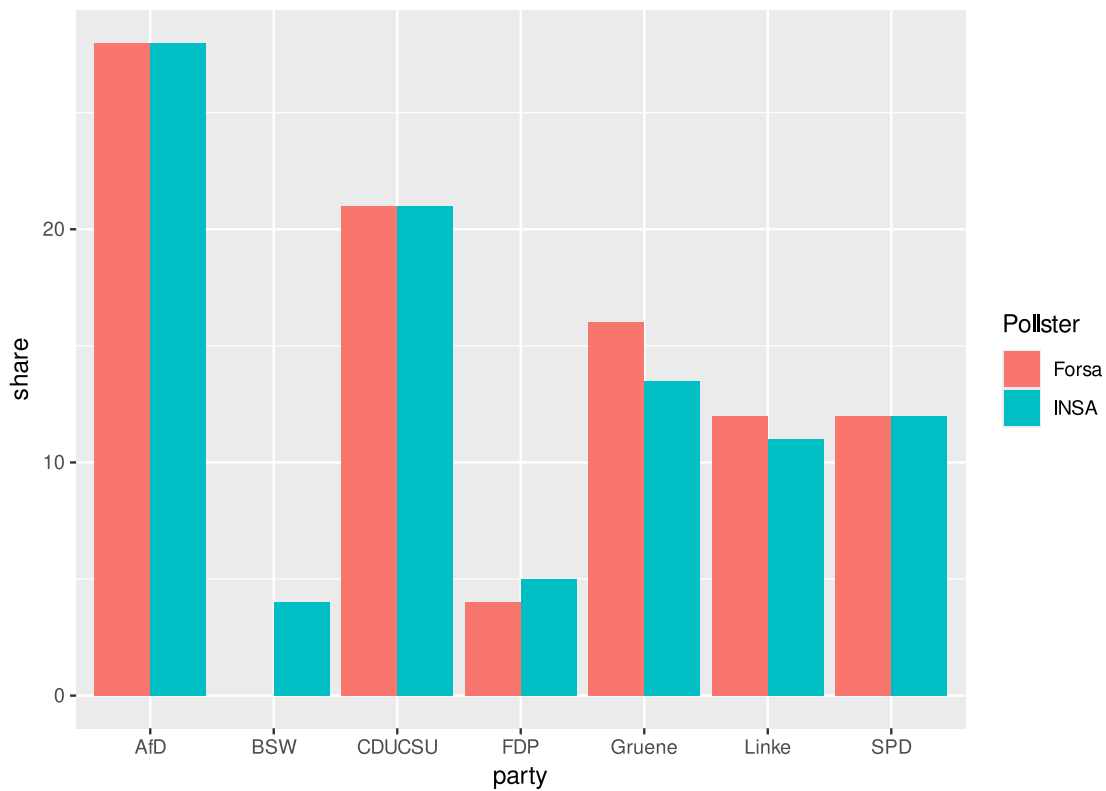
Groups

A second mapped variable (color, fill, or group) clusters bars side by side automatically.

```
polls |>
  filter(Date == max(Date)) |>
  ggplot(aes(x = party, y = share, fill = Pollster)) +
  geom_col(position = "dodge")
```

⚠ WARNING

Warning: Removed 1 row containing missing values or values outside the scale range (`geom\_col()`).

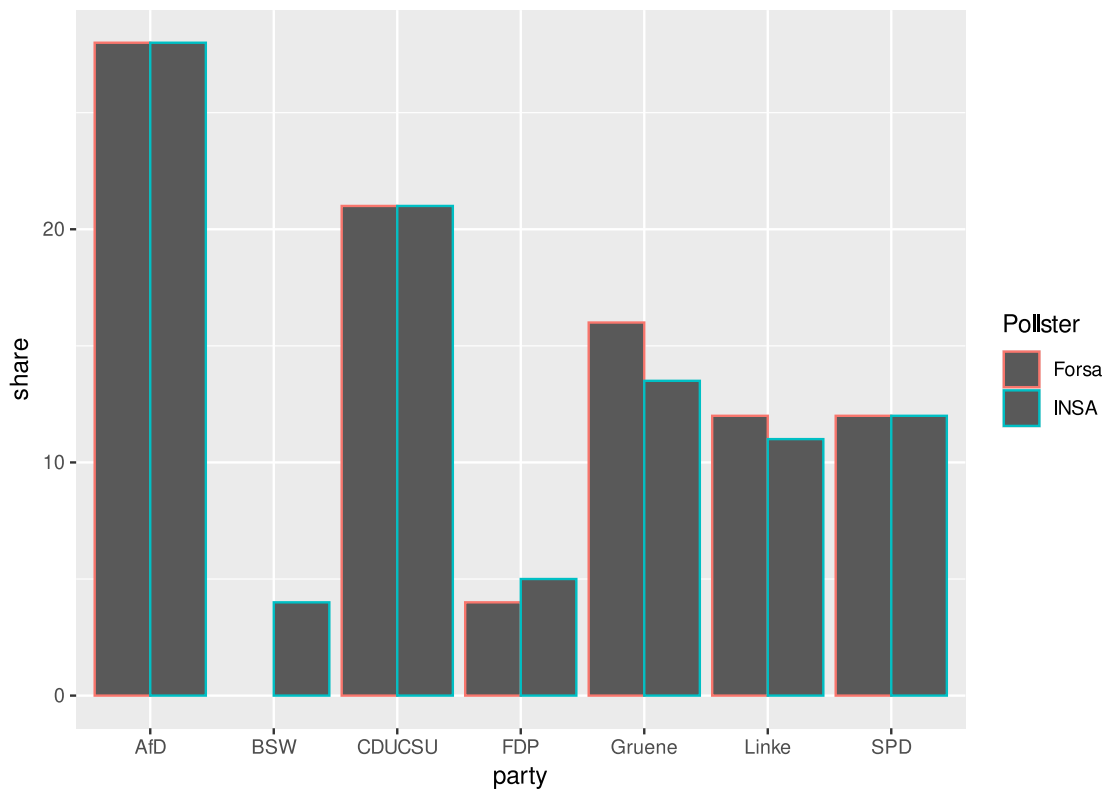


color only outlines the bars — for `geom_col()` you almost always want `fill`.

```
polls |>
  filter(Date == max(Date)) |>
  ggplot(aes(x = party, y = share, color = Pollster)) +
  geom_col(position = "dodge")
```

⚠ WARNING

Warning: Removed 1 row containing missing values or values outside the scale range (``geom_col()``).

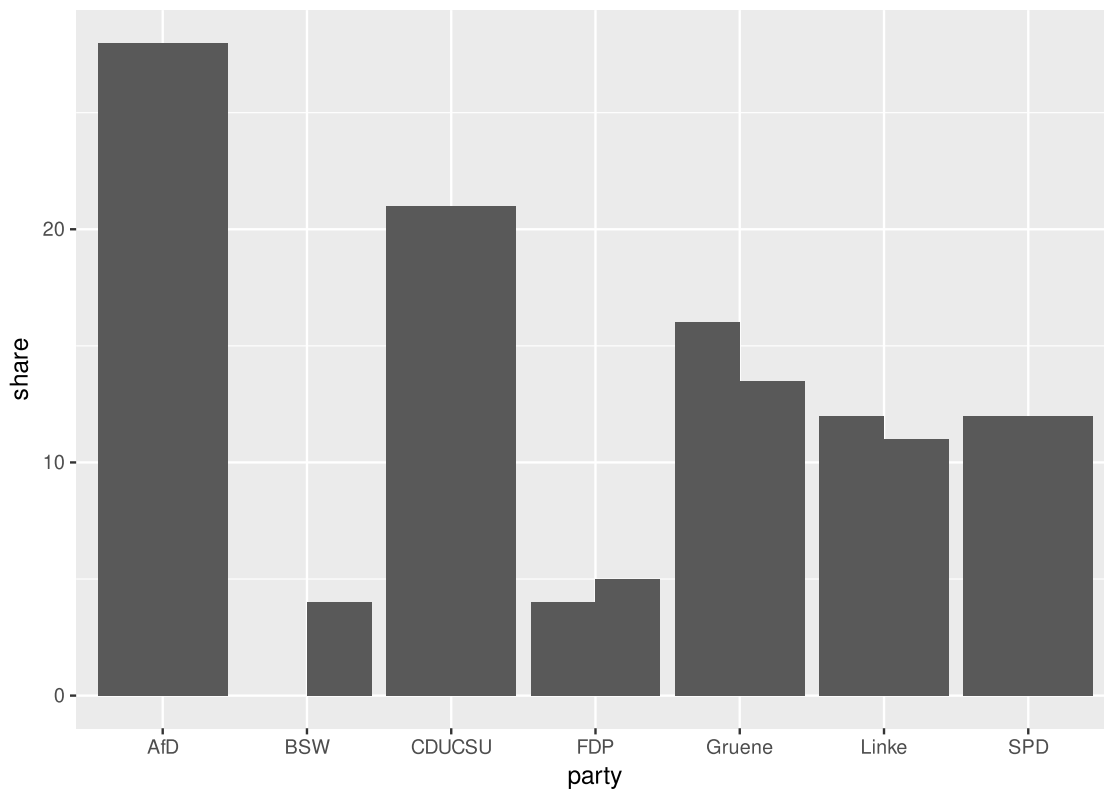


group alone still dodges the bars into position, just without any visual distinction between them.

```
polls |>
  filter(Date == max(Date)) |>
  ggplot(aes(x = party, y = share, group = Pollster)) +
  geom_col(position = "dodge")
```

⚠ WARNING

Warning: Removed 1 row containing missing values or values outside the scale range (`geom\_col()`).



Color/fill and group don't have to match — group decides which points `geom_line()` connects, independently of what color encodes.

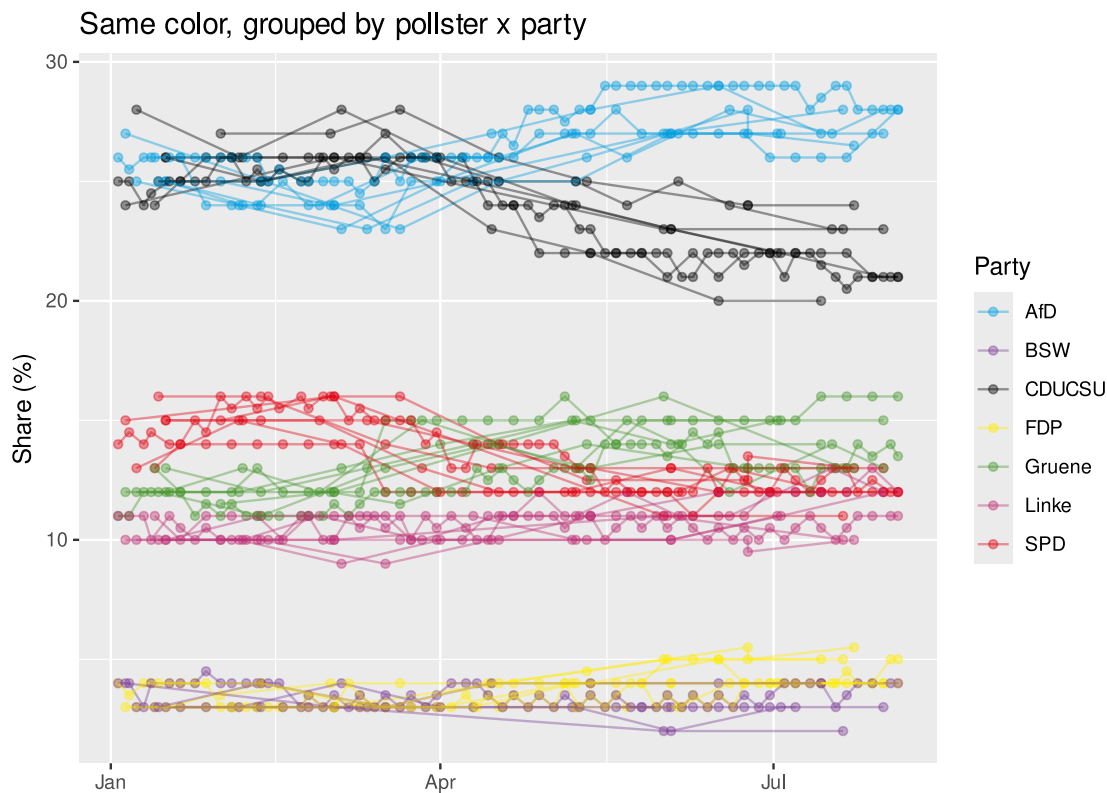
```
polls |>
  filter(between(Date, as.Date("2026-01-01"), Sys.Date())) |>
  ggplot(aes(x = Date, y = share, color = party, group = paste(Pollster, party))) +
  geom_point(alpha = 0.4) +
  geom_line(alpha = 0.4) +
  scale_color_manual(values = party_colors) +
  labs(title = "Same color, grouped by pollster x party", x = NULL, y = "Share (%)",
  color = "Party")
```

⚠ WARNING

Warning: Removed 39 rows containing missing values or values outside the scale range (``geom_point()``).

⚠ WARNING

Warning: Removed 28 rows containing missing values or values outside the scale range (``geom_line()``).



Faceting

One small panel per pollster instead of one crowded plot.

```
afd |>
  ggplot(aes(x = Date, y = share)) +
  geom_point(alpha = 0.4) +
  geom_smooth(span = .4, se = FALSE) +
  facet_wrap(~Pollster)
```

MESSAGE

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

WARNING

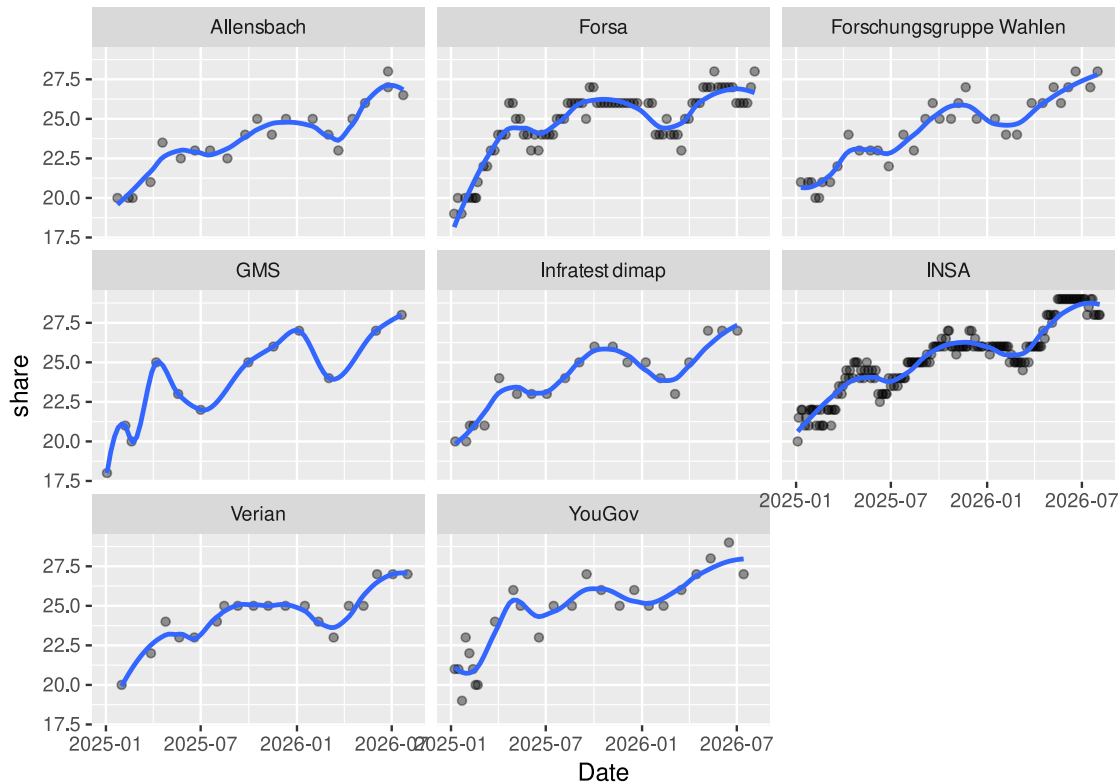
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 20138

WARNING

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 47

⚠ WARNING

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 0



Facet by party instead — `scales = "free_y"` lets each panel use its own y-axis range, since parties poll at very different levels.

```
polls |>
  filter(Pollster == "Forsa") |>
  ggplot(aes(x = Date, y = share, color = party)) +
  geom_point(alpha = 0.05) +
  geom_smooth(span = .4, se = FALSE) +
  scale_color_manual(values = party_colors) +
  facet_wrap(~party, scales = "free_y")
```

📄 MESSAGE

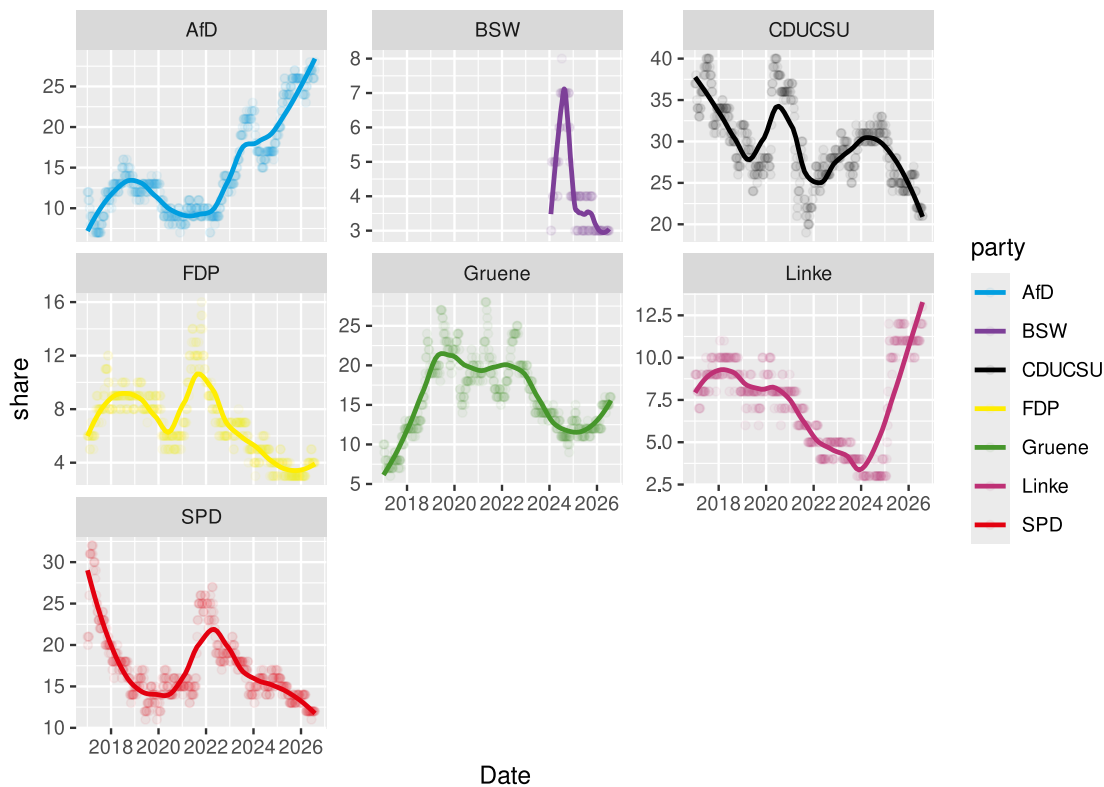
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

⚠ WARNING

Warning: Removed 397 rows containing non-finite outside the scale range
(`stat\_smooth()`).

⚠ WARNING

Warning: Removed 397 rows containing missing values or values outside the scale range
(`geom\_point()`).



Try it: the full party landscape

Your task: combine color scales, faceting, and labels into one plot — a trend line per party, real party colors, one panel per pollster.

```
polls |>
  ggplot(aes(x = Date, y = share, color = party)) +
  geom_point(alpha = .05) +
  geom_smooth(span = .4, se = FALSE) +
  scale_color_manual(values = party_colors) +
  facet_wrap(~Pollster) +
  labs(title = "Bundestag polling trend by pollster", x = "", y = "Share (%)", color
       = "Party")
```

MESSAGE

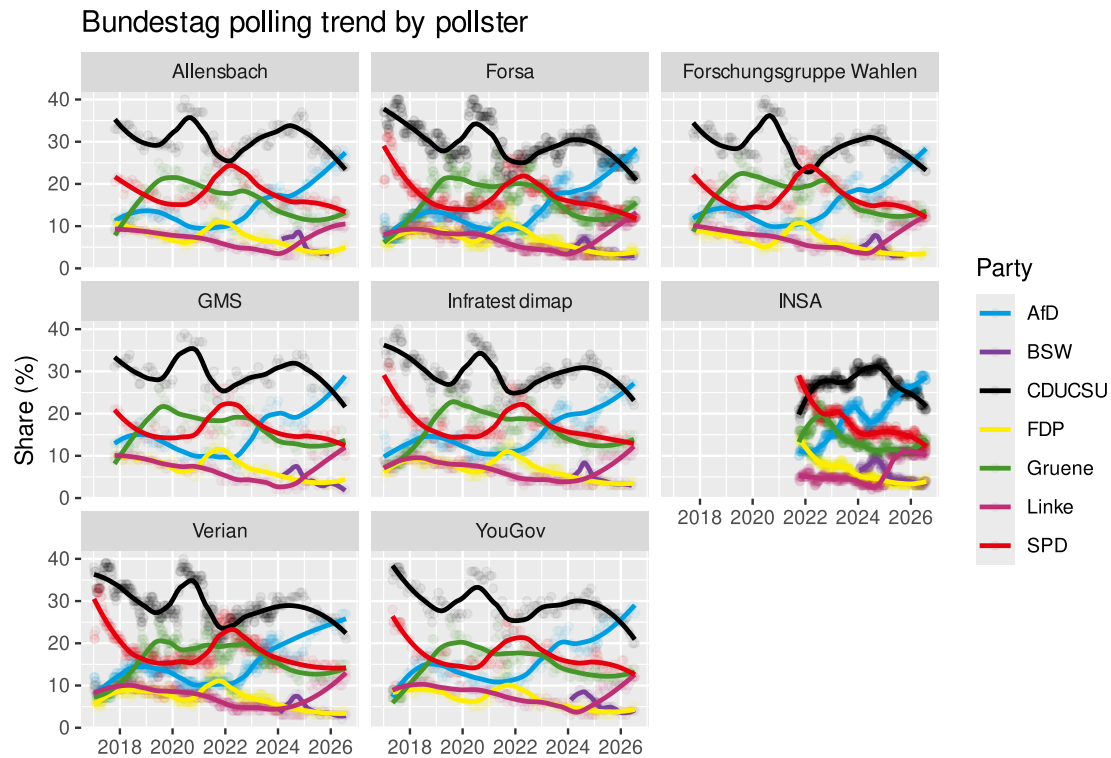
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

WARNING

Warning: Removed 1548 rows containing non-finite outside the scale range (``stat_smooth()``).

⚠ WARNING

Warning: Removed 1548 rows containing missing values or values outside the scale range
(`geom\_point()`).



Try it: how each party has moved since 2017

Your task: show how each party's Forsa-pollled support has shifted between its very first poll and the most recent one. No new geom needed — think about what a plot connecting just two points per party would look like.

```
forsa_first_last <- polls |>
  filter(Pollster == "Forsa") |>
  filter(Date %in% c(min(Date), max(Date)))
forsa_first_last
```

> OUTPUT

```
# A tibble: 14 × 7
  Poll_ID Pollster Date      n Period      party share
  <chr>    <chr>   <date>   <dbl> <chr>      <chr>   <dbl>
1 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. CDU/CSU    37
2 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. SPD      21
3 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. Gruene   10
4 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. FDP       6
5 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. Linke     9
6 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. AfD      12
7 BT021701040 Forsa 2017-01-04 2001 27.12.-30.12. BSW      NA
8 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. CDU/CSU    21
9 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. SPD      12
10 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. Gruene   16
11 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. FDP       4
12 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. Linke    12
13 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. AfD      28
14 BT02260804Q Forsa 2026-08-04 2501 28.07.-03.08. BSW      NA
```

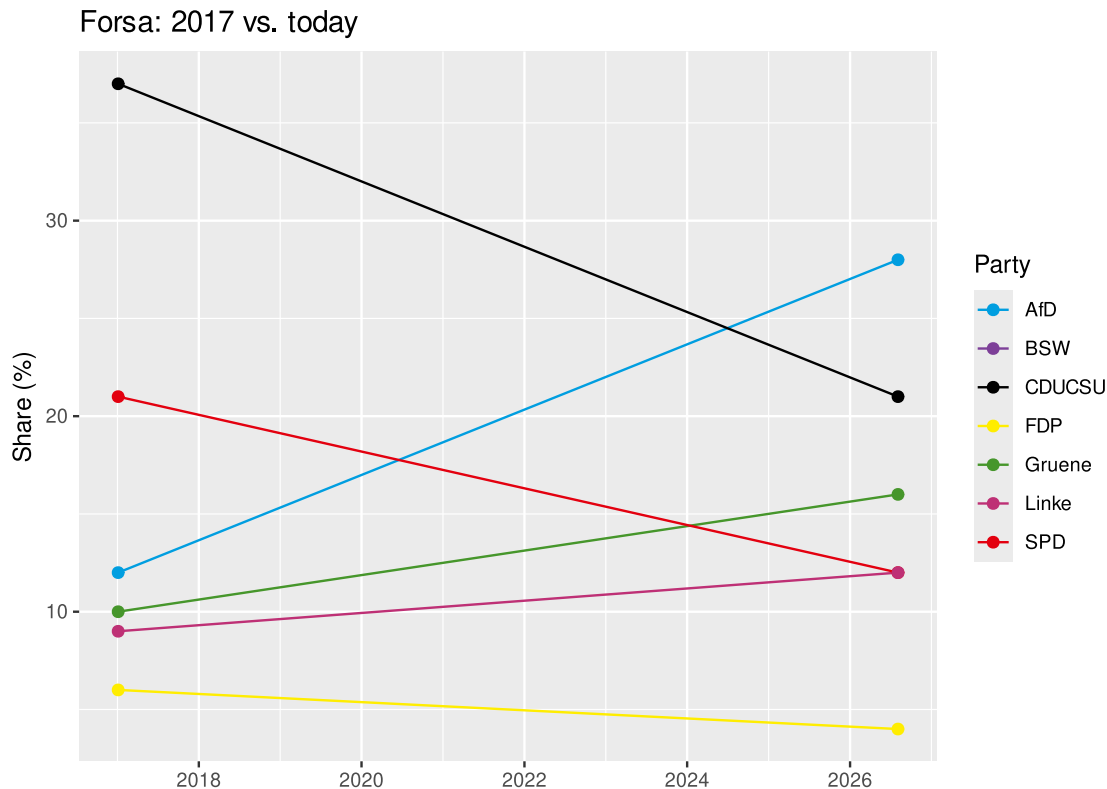
```
forsa_first_last |>
  ggplot(aes(x = Date, y = share, color = party)) +
  geom_line() +
  geom_point(size = 2) +
  scale_color_manual(values = party_colors) +
  labs(title = "Forsa: 2017 vs. today", x = NULL, y = "Share (%)", color = "Party")
```

⚠ WARNING

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_line()`).

⚠ WARNING

Warning: Removed 2 rows containing missing values or values outside the scale range (`geom\_point()`).



Try it: did BSW clear the 5% threshold?

Your task: plot BSW's Forsa-pollled trend over time and show whether — and when — it crosses the 5% electoral threshold. A reference line at 5% helps make the answer visible at a glance.

```
bsw_forsa <- polls |>
  filter(Pollster == "Forsa", party == "BSW") |>
  filter(!is.na(share))
bsw_forsa
```

> OUTPUT

```
# A tibble: 121 x 7
  Poll_ID Pollster Date      n Period      party share
  <chr>    <chr>   <date>  <dbl> <chr>    <chr> <dbl>
1 BT02240123K Forsa 2024-01-23 2503 16.01.-22.01. BSW      3
2 BT02240130L Forsa 2024-01-30 2506 23.01.-29.01. BSW      3
3 BT02240206M Forsa 2024-02-06 2503 30.01.-04.02. BSW      5
4 BT02240213N Forsa 2024-02-13 2502 06.02.-12.02. BSW      5
5 BT02240220O Forsa 2024-02-20 2504 13.02.-19.02. BSW      4
6 BT02240227P Forsa 2024-02-27 2503 20.02.-26.02. BSW      4
7 BT02240305Q Forsa 2024-03-05 2506 27.02.-04.03. BSW      4
8 BT02240312R Forsa 2024-03-12 2502 05.03.-11.03. BSW      5
9 BT02240319S Forsa 2024-03-19 2500 12.03.-18.03. BSW      5
10 BT02240326T Forsa 2024-03-26 2503 19.03.-25.03. BSW      5
# i 111 more rows
```

```
bsw_forsa |>
  ggplot(aes(x = Date, y = share, color = party)) +
  geom_point(alpha = 0.4) +
  geom_smooth(span = .5, se = FALSE) +
  geom_hline(yintercept = 5, linetype = "dashed") +
  scale_color_manual(values = party_colors) +
  labs(title = "BSW vs. the 5% threshold (Forsa)", x = NULL, y = "Share (%)")
```

MESSAGE

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

